Guidance and Control of Unmanned Autonomous Vehicles

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Lateral Guidance Strategy for Straight-line Following

- For $\chi^\infty \in (0, \pi/2]$, $\boxed{-\frac{\pi}{2} \leq -\chi^\infty \frac{2}{\pi} \tan^{-1}\left(k_{\text{path}} e_{py}\right) \leq \frac{\pi}{2}}$
- Commanded course of the vehicle for straight-line tracking

$$\boxed{\chi^c = \chi_q + \chi_d(e_{py}) = \chi_q - \chi^\infty \frac{2}{\pi} \tan^{-1}\left(k_{\text{path}} e_{py}\right)}$$

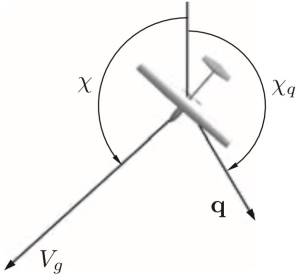
- Lyapunov function candidate: $\mathcal{V}(e_{py}) = \frac{1}{2} e_{py}^2$
- Assume that course controller works properly and thus $\boxed{\chi = \chi_q + \chi_d(e_{py})}$
- Derivative of the Lyapunov function candidate $\dot{\mathcal{V}} = \dot{e}_{py} e_{py}$

$$\dot{\mathcal{V}} = V_g \sin\left(\chi - \chi_q\right) e_{py} = -V_g e_{py} \sin\left\{\chi^\infty \frac{2}{\pi} \tan^{-1}\left(k_{\text{path}} e_{py}\right)\right\} \cos\gamma < \ 0 \ \forall \ e_{py} \neq 0$$

- As a result, $e_{py} \to 0$ asymptotically.

Lateral Guidance Strategy for Straight-line Following

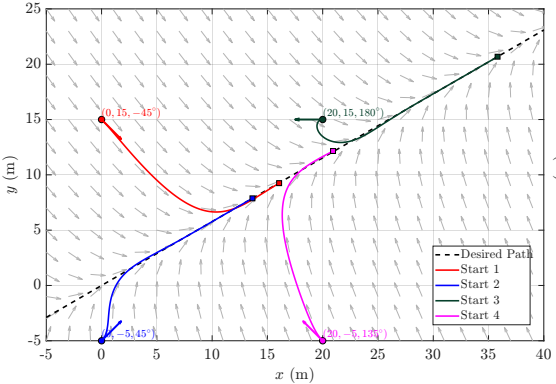
- Commanded course may lead to undesirable behavior due to range of function atan2 in $\pm\pi$
- Assume χ_q little smaller than π.
- For χ negative, it turns right to make corrections and align with the desired path.
- Not a suitable maneuver.

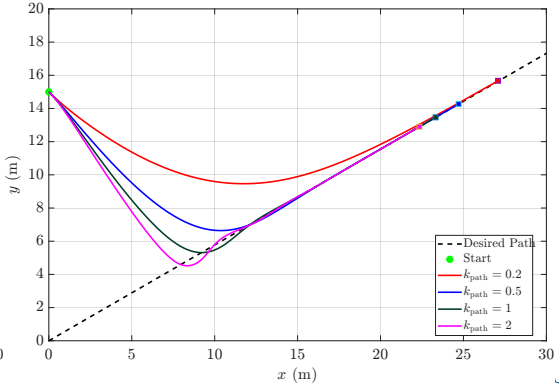


- What would be possible solutions to avoid such issues?
- Express χ_q as a negative quantity less than $-\pi$, and then the UAV will turn left to align with the desired course. With $m \in \mathcal{N}$,

$$\chi_q = \tan^{-1}\left(\frac{q_e}{q_n}\right) + 2\pi m \quad \text{s.t.} \quad -\pi \leq \chi_q - \chi \leq \pi$$

Lateral Guidance Strategy for Straight-line Following



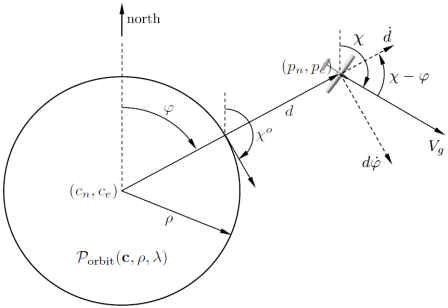


Orbit Path Following

How to specify an orbit path to be tracked by the UAV?

Orbit Definition

$$\mathcal{P}_{\text{orbit}}(\boldsymbol{c}, \rho, \lambda) = \left\{ \boldsymbol{r} \in \mathbb{R}^3 : \boldsymbol{r} = \boldsymbol{c} + \lambda\rho\,(\cos\varphi,\ \sin\varphi,\ 0)^T, \varphi \in [0, 2\pi) \right\}$$



- Center: $\boldsymbol{c} \in \mathbb{R}^3$, Radius: $\rho \in \mathbb{R}$
- Direction: $\lambda = 1$ (CW); $\lambda = -1$ (CCW)
- **Assumption**: Center of the orbit expressed in inertial coordinates
- d: Radial distance from the desired center of the orbit to the UAV
- φ: Phase angle of relative position of UAV

Orbit Path Following

- How to obtain a guidance strategy for following an orbit path by the vehicle?
- In polar coordinates, radial position of vehicle from center of orbit and phase angle of relative position

$$d = \sqrt{(p_n - c_n)^2 + (p_e - c_e)^2}, \quad \varphi = \tan^{-1}\left[\frac{p_e - c_e}{p_n - c_n}\right]$$

- Constant-altitude UAV dynamics in the North and East directions

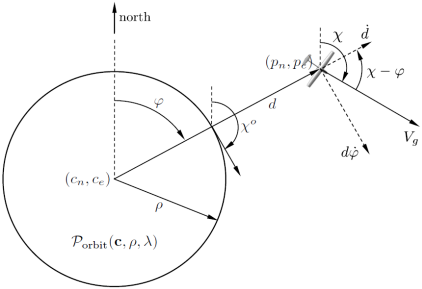
$$\begin{bmatrix} \dot{p}_n \\ \dot{p}_e \end{bmatrix} = \begin{bmatrix} V_g \cos\chi \\ V_g \sin\chi \end{bmatrix}$$

- In polar coordinates, equations of motion of UAV

$$\begin{bmatrix} \dot{d} \\ d\dot{\varphi} \end{bmatrix} = \begin{bmatrix} \cos\varphi & \sin\varphi \\ -\sin\varphi & \cos\varphi \end{bmatrix} \begin{bmatrix} \dot{p}_n \\ \dot{p}_e \end{bmatrix} = \begin{bmatrix} \cos\varphi & \sin\varphi \\ -\sin\varphi & \cos\varphi \end{bmatrix} \begin{bmatrix} V_g \cos\chi \\ V_g \sin\chi \end{bmatrix} = \begin{bmatrix} V_g \cos(\chi - \varphi) \\ V_g \sin(\chi - \varphi) \end{bmatrix}$$

Orbit Path Following

Can we also obtain the same vehicle dynamics using the geometry?



Vehicle dynamics in polar coordinates

$$\dot{d} = V_g \cos(\chi - \varphi)$$

$$\dot{\varphi} = \frac{V_g}{d} \sin(\chi - \varphi)$$

$$\ddot{\chi} = -b_{\dot{\chi}}\dot{\chi} + b_{\chi}(\chi^c - \chi)$$

- How to obtain the desired course angle of vehicle, χ^o?
- For clockwise orbit, desired course angle $\chi^o = \varphi + \frac{\pi}{2}$
- For counterclockwise orbit, desired course angle, $\chi^o = \varphi - \frac{\pi}{2}$

$$\boxed{\chi^o = \varphi + \lambda\frac{\pi}{2}}$$

Orbit Path Following

- **Control objectives**: To drive $d(t) \to \rho$ and $\chi(t) \to \chi^o$ in presence of wind.
- Can we use a similar idea as that for the straight-line path following?
- Yes, using the desired vector field to make the vehicle follow an orbit
- For large $d \gg \rho$, the UAV should fly toward the center of orbit

$$\boxed{\chi_d \approx \chi^0 + \lambda \frac{\pi}{2}}$$

- For $d = \rho$, $\boxed{\chi_d = \chi^0}$
- Candidate course field of the UAV, valid for all $d \geq 0$

$$\boxed{\chi_d(d-\rho, \lambda) = \chi^0 + \lambda \tan^{-1}\left\{k_{\text{orbit}}\left(\frac{d-\rho}{\rho}\right)\right\}, \quad k_{\text{orbit}} > 0}$$

- Parameter k_{orbit} specifies the rate of transition from $\lambda\pi/2$ to zero.
- How to guarantee the stability of UAV with this controller?

Orbit Path Following

- Lyapunov function candidate: $\mathcal{V} = \frac{1}{2}(d-\rho)^2$
- Derivative of the Lyapunov function candidate, with $\chi = \chi_d$,

$$\dot{\mathcal{V}} = (d-\rho)\, V_g \cos(\chi - \varphi) = (d-\rho)\, V_g \cos(\chi_d - \varphi)$$

- Using the chosen value of χ_d

$$\begin{aligned}\dot{\mathcal{V}} =& V_g (d-\rho) \cos\left[\chi^0 + \lambda \tan^{-1}\left\{k_{\text{orbit}}\left(\frac{d-\rho}{\rho}\right)\right\} - \varphi\right] \\ =& V_g (d-\rho) \cos\left[\varphi + \lambda\frac{\pi}{2} + \lambda \tan^{-1}\left\{k_{\text{orbit}}\left(\frac{d-\rho}{\rho}\right)\right\} - \varphi\right] \\ =& -V_g (d-\rho) \sin\left[\tan^{-1}\left\{k_{\text{orbit}}\left(\frac{d-\rho}{\rho}\right)\right\}\right] < 0 \forall\ d \neq \rho\end{aligned}$$

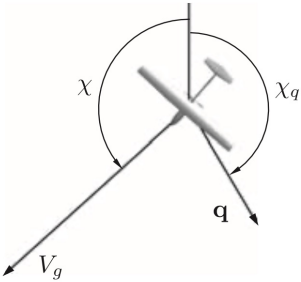
- As a result, $d \to \rho$ asymptotically.

Orbit Path Following

- Course command for UAV to follow orbit path, using $\chi^o = \varphi + \lambda \frac{\pi}{2}$

$$\chi^c(t) = \chi_d = \varphi + \lambda \left[\frac{\pi}{2} + \tan^{-1} \left\{ k_{\text{orbit}} \left(\frac{d - \rho}{\rho} \right) \right\} \right]$$

- As before, for the case of χ_q, computation of φ should be proper to avoid a sudden jump in course angle when φ changes from $-\pi$ to π.

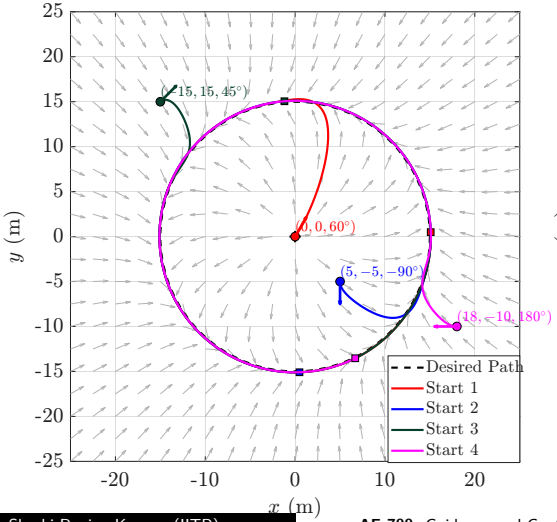


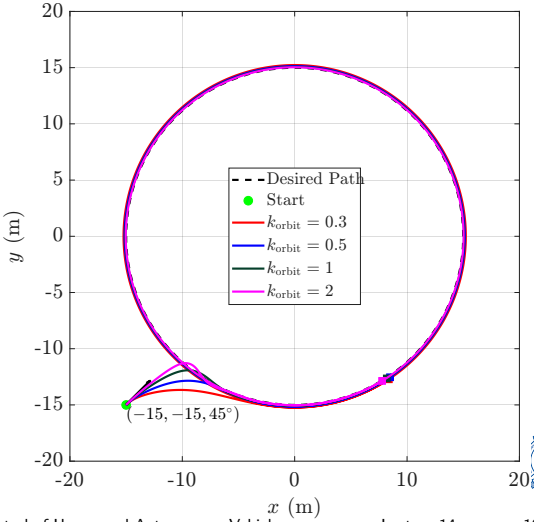
To avoid such issue, φ should be computed as

$$\varphi = \tan^{-1} \left(\frac{p_e - c_e}{p_n - c_n} \right) + 2\pi m$$

with $m \in \mathcal{N}$ chosen such that $-\pi \leq \varphi - \chi \leq \pi$.

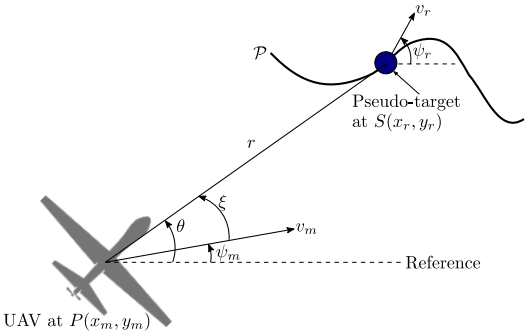
Orbit Path Following





Classification of path-following algorithms

- **Geometric methods**
  - ⋆ Virtual target-based approach
    - ▶ Pure pursuit
    - ▶ Line-of-sight (LOS)
    - ▶ Nonlinear guidance law (NLGL)
  - ⋆ Dubin's curves-based approach
  - ⋆ Vector-field-based approach



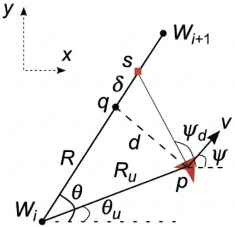
- **Virtual target point (VTP)**: Guidance laws direct UAVs to chase VTP, which eventually drives vehicles onto the path.
- **Virtual distance**: Distance between VTP and UAV position projected on the path
- **Control techniques**: PID, LQR, SMC, Backstepping, Adaptive, etc.

Carrot-chasing algorithm

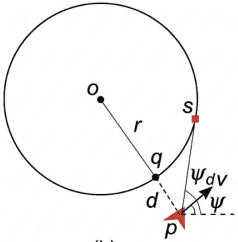
- Select a VTP on the desired path
- UAV updates its heading direction toward VTP
- VTP is referred to as the carrot
- UAV chases the carrot

Straight-line following

- Define a path using two waypoints W_i and W_{i+1}
- Calculate path track error d
- Update location of VTP
- Update the heading angle ψ and speed$u$

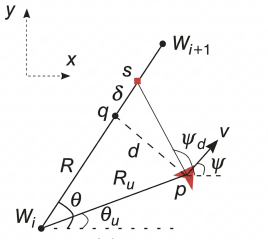


(a)



(b)

Carrot-chasing algorithm: Straight-line



ALGORITHM 2 Algorithm for straight-line following using the carrot-chasing algorithm.

1: Initialize: $W_i = (x_i, y_i), W_{i+1} = (x_{i+1}, y_{i+1}), p = (x, y), \psi, \delta, v_a, \kappa$

2: $R_u = \| W_i - p \|, \theta = \text{atan2}(y_{i+1} - y_i, x_{i+1} - x_i)$

3: $\theta_u = \text{atan2}(y - y_i, x - x_i), \beta = \theta - \theta_u$

4: $R = \sqrt{R_u^2 - (R_u \sin(\beta))^2}$

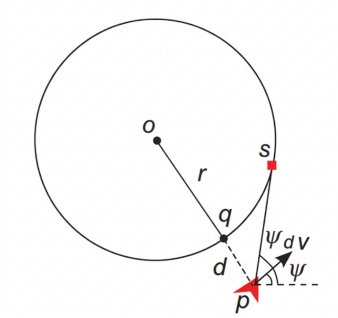
5: $(x_t, y_t) \leftarrow ((R + \delta)\cos\theta, (R + \delta)\sin\theta)$ % $s = (x_t, y_t)$

6: $\psi_d = \text{atan2}(y_t - y, x_t - x)$

7: $u = \text{max_limit}(\kappa(\psi_d - \psi) v_a)$

- Consider UAV's is at a location p with heading angle ψ
- q : Projection of p onto LOS
- R : distance of q from W_i
- d : cross-track error
- Assume a VTP $s = (x_t, y_t)$ at a distance δ from q
- Update ψ_d based on location of s using Algorithm 2
- Continue until the UAV reaches W_{i+1}

Carrot-chasing algorithm: Loiter path



ALGORITHM 3 Algorithm to loiter using the carrot-chasing algorithm.

1: Initialize: $O = (x_l, y_l), r, p, \psi, \lambda, v_a, \kappa$

2: $d = \|O - p\| - r$

3: $\theta = \text{atan2}(y - y_l, x - x_l)$

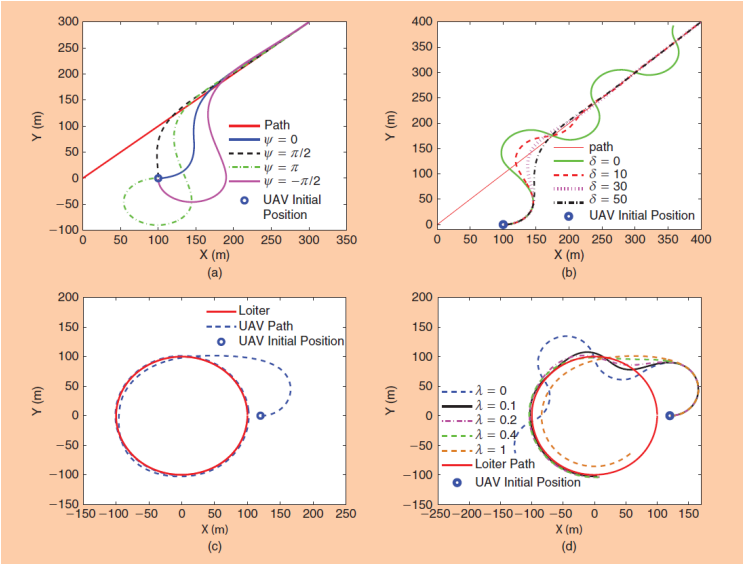
4: $(x_t, y_t) \leftarrow (r\cos(\theta + \lambda), r\sin(\theta + \lambda))$ % $s = (x_t, y_t)$

5: $\psi_d = \text{atan2}(y_t - y, x_t - x)$

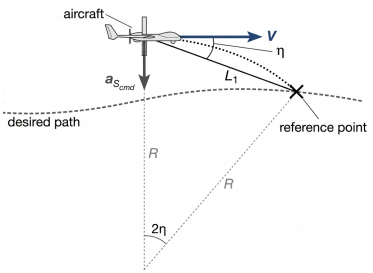
6: $u = \text{max_limit}(\kappa(\psi_d - \psi)v_a)$

- **Objective**: UAV has to circle around a point with a standoff distance of r
- How to follow the loiter path?
- UAV must be located on the circle
- Heading direction $\perp$ $\boldsymbol{r}$
- p : Initial location of UAV
- Cross track error $d = \|O - p\| - r$
- Assume a VTP, s, at an angular displacement of λ from q
- Update UAV and VTP using Algorithm 3

Simulation result



NLGL based on pure pursuit



- A reference point on the desired path chosen at a constant distance (L_1) forward of the vehicle.
- Lateral acceleration command generated based on direction of reference point, relative to UAV's velocity.
- $a_{s_{\text{cmd}}}$: Steering control of UAV

- How to compute necessary UAV acceleration to follow the above-curved path?

$$L_1 = 2R\sin\eta \implies \sin\eta = \frac{L_1}{2R} \implies \boxed{R = \frac{L_1}{2\sin\eta}}$$

- **Centripetal acceleration**: $\boxed{a_{s_{\text{cmd}}} = \frac{V^2}{R} = \frac{2V^2}{L_1}\sin\eta}$

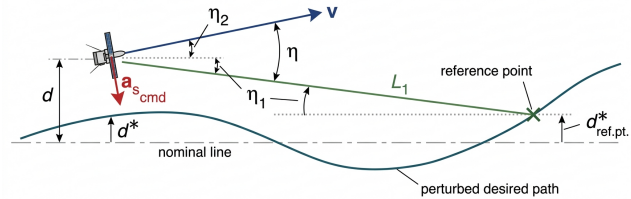
NLGL based on pure pursuit

What are your observations on this path following method?

- **Direction of acceleration**: Sign of the angle between the line segment L_1 and vehicle velocity vector, look for expression to support the same
- **Vehicle far from desired path**: Guidance law tends to rotate the velocity direction to approach the desired path at a large angle
- **Vehicle close to desired path**: Guidance law rotates the velocity direction to approach the desired path at a small angle
- A circle can be drawn to pass through the UAV and the reference point, and the UAV's velocity vector tangential to the circle.

NLGL

- How to choose the look-ahead distance L_1 in this approach?
- Can we take any arbitrary value? What would be the consequences?



- **Desired path**: A small perturbation from straight-line
- L_1: Distance between vehicle and reference point on the desired path
- V : Ground speed
- $d, d^\star$, and $d^\star_{\text{ref.pt.}}$: Lateral perturbations of UAV, desired path, and reference point from nominal straight line, respectively

NLGL

- Assuming magnitude of η is small, one can write $\boxed{\sin\eta \approx \eta = \eta_1 + \eta_2}$
- How can we obtain the angles η_1 and η_2?

$$\boxed{\eta_1 = \frac{d - d^{\star}_{\text{ref.pt.}}}{L_1}, \quad \eta_2 = \frac{\dot{d}}{V}}$$

- For a straight-line case ($d^{\star}_{\text{ref.pt.}} = 0$)

$$\boxed{a_{s_{\text{cmd}}} = \frac{2V^2}{L_1}\sin\eta \approx \frac{2V^2}{L_1}\eta = \frac{2V^2}{L_1}\left(\frac{d - d^{\star}_{\text{ref.pt.}}}{L_1} + \frac{\dot{d}}{V}\right) = 2\frac{V}{L_1}\left(\dot{d} + \frac{V}{L_1}d\right)}$$

- Any observations? PD control structure of the acceleration command
- **Controller gains**: UAV's speed V and look-ahead distance L_1
- Small value for L_1 leads to a high controller gain
- **Time constant of the controller**: L_1/V

NLGL

- How to select look-ahead distance L_1 for this controller design?
- Perform stability analysis with a linear plant model and derive a linear controller
- Plant model should include UAV dynamics with inner-loop bank angle controller (if bank angle is used to generate lateral acceleration for the aircraft)
- Include sensor dynamics in the associated loop transmission function
- For general tracking problem with perturbed trajectory, $d^{\star}_{\text{ref.pt.}} \neq 0$
- For small perturbations, $\ddot{d} = -a_{s_{\text{cmd}}}$

$$\ddot{d} = -a_{s_{\text{cmd}}} = -\frac{V^2}{L_1}\left(\frac{d - d^{\star}_{\text{ref.pt.}}}{L_1} + \frac{\dot{d}}{V}\right) \implies \boxed{\ddot{d} + \frac{2V}{L_1}\dot{d} + 2\frac{V^2}{L_1^2}d = \frac{2V^2}{L_1^2}d^{\star}_{\text{ref.pt.}}}$$

NLGL

- Using the Laplace transform, one can obtain

$$\boxed{\frac{d(s)}{d^{\star}_{\text{ref.pt.}}(s)} = \frac{2(V/L_1)^2}{s^2 + (2V/L_1)s + 2(V/L_1)^2}}$$

- Can we obtain undamped natural frequency and damping ratio?
- Damping ratio $\zeta = 0.707$ and undamped natural frequency $\omega_n = \sqrt{2}V/L_1$
- What is the input in the above transfer function here?
- **Lateral position of reference point**, Not position of desired path at current UAV location ($d^{\star}_{\text{ref.pt.}} \neq d^{\star}$)
- **Guidance law input comes from a reference point that is at "look-ahead" distance L_1 ahead on the desired path**

NLGL

- Use of look-ahead distance helpful in the phase recovery
- Assume that angle η is small
- Time difference between $d^\star$ and $d^\star_{\text{ref.pt.}}$: L_1/V

$$\frac{d^\star_{\text{ref.pt.}}(s)}{d^\star(s)} \approx e^{\tau s}, \quad \tau = \frac{L_1}{V}$$

- Transfer function from $d^\star$ to d

$$\frac{d(s)}{d^\star(s)} = \frac{\omega_n^2 e^{\tau s}}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

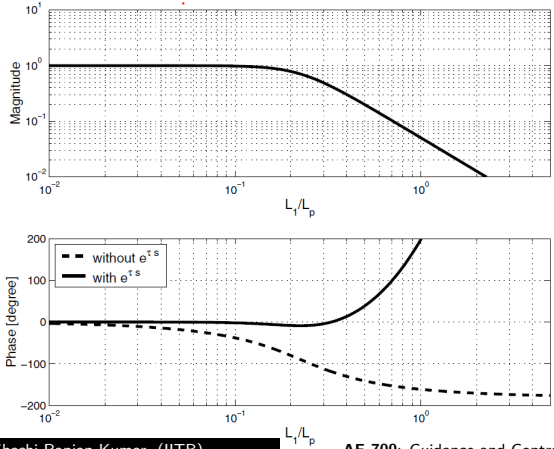
- Consider a sinusoidal trajectory command

$$d^\star = A \sin\left(\frac{2\pi x}{L_p}\right)$$

- A : Small path amplitude
- L_p: Spatial wavelength of the input sinusoid
- Smaller values of L_p imply higher spatial frequencies
- x : Distance along the desired path

NLGL

- Assume $x \approx Vt$ implies an input frequency $2\pi V/L_p$
- Undamped natural frequency $\omega_n = \sqrt{2}V/L_1$



- **Wavelength of input**:

$$L_p = \sqrt{2}\pi L_1 \approx 4.4 L_1$$

- $d^\star$ excites system at

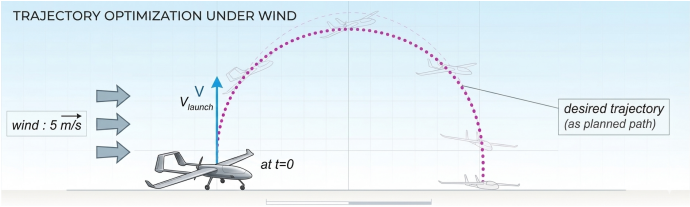
$$2\pi V/L_p = 2\pi V/(\sqrt{2}\pi L_1)$$
$$= \sqrt{2}V/L_1 = \omega_n$$

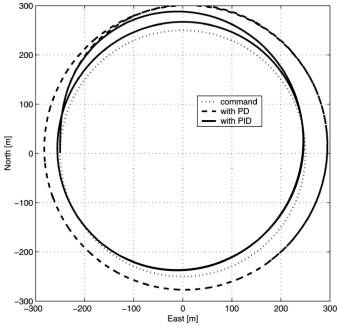
- **Phase lag**: 90° at ω_n
- Phase lag is referenced from position input of reference point $d^\star_{\text{ref.pt.}}$, not from $d^\star$

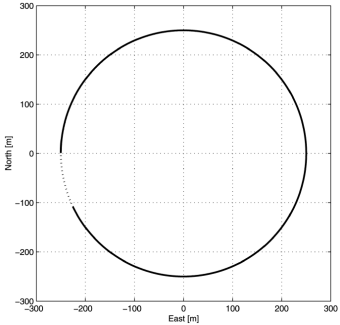
NLGL

- Reference point at $L_1 = L_p/4.4$, Quarter period distance ahead
- Phase lead in $d^\star_{\text{ref.pt.}}$ over $d^\star$: 90°
- Two effects tend to cancel each other
- Phase difference between d and $d^\star$ reduced significantly.
- Can you observe the same from the Bode plot?
- If L_p is the wavelength of the highest frequency content in the desired path, then look-ahead distance L_1 must be chosen to be less than about $L_p/4.4$.
- **Enables phase recovery around the bandwidth frequency** ω_n
- **Small perturbations about a straight line path**: NLGL approximates a linear PD controller operating on cross-track error.
- What about the performance with the loiter path with wind?

PD vs NLGL: Performance with loiter path with wind







- Vehicle speed: 25 m/s
- Look-ahead distance

$$L_1 = 150 \text{ m}$$

- Steady-state wind: 5 m/s (West to East)
- Initial heading of UAV: North
- PD controller: Cross-track error range 30-60 m
- PID controller: Cross-track error ± 20 m
- Any other observations?

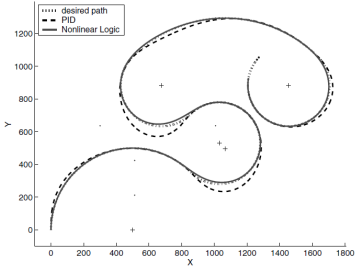
PD Vs NLGL

- **Downwind region**: UAV flies outside the circular path
- **Upwind region**: UAV flies inside the circular path
- Any possible reasons you can think of?
- In downwind region, UAV moves faster with respect to the inertial frame.
- UAV must generate larger acceleration command (or larger bank angle command) Why?
- Linear feedback controller with fixed gain cannot remove the error immediately
- NLGL performed well, Why?
- NLGL uses the ground speed V to generate the acceleration command

PD Vs NLGL

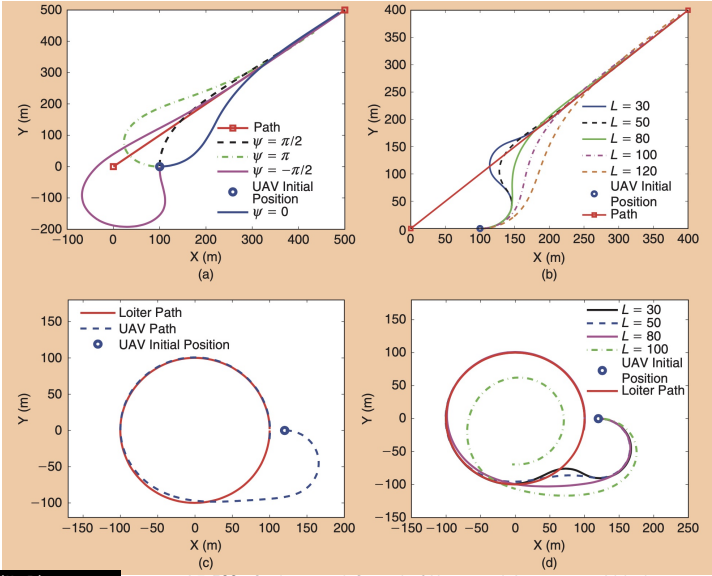
Linked Circular Path Following

- Path composed of five circular segments with a minimum radius of 250 m
- $L_1 = 0.6R$ for the second, third, and fifth segments
- PID: Cross-track error 70 m for the second and third segments
- NLGL: Cross-track error 15 m for the same



- Can we improve the accuracy of PID?
- Use PID with gain-scheduling and accounting for the ground speed
- High error during the initial transient period
- For commanded path with many changes of curvature, PID suffers

Control Architecture



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Thank you for your attention !!!